IoT MINI PROJECT · PHASE 1: PROJECT PROPOSAL

# Addressing Class Imbalance in CICIoT2023-Based Intrusion Detection via SMOTE and Feature Selection



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Base paper: E. C. P. Neto et al., “CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment,” Sensors 23(13):5941, 2023

# High accuracy can still miss the attacks that matter

## ≈ 2,600 : 1

DDoS flows for every Brute-Force flow in CICIoT2023

*A model that always answers "DDoS" is already ≈ 73% accurate, while catching zero brute-force or web attacks.*



### Rare is not harmless

Brute-force logins and web injections are how IoT devices get taken over, not just knocked offline.



### Accuracy hides the failure

DDoS and DoS make up ≈ 90% of the flows, so they dominate accuracy and weighted scores.



### Per-class recall is the honest test

This project measures and improves recall on the rare classes, not just the headline number.

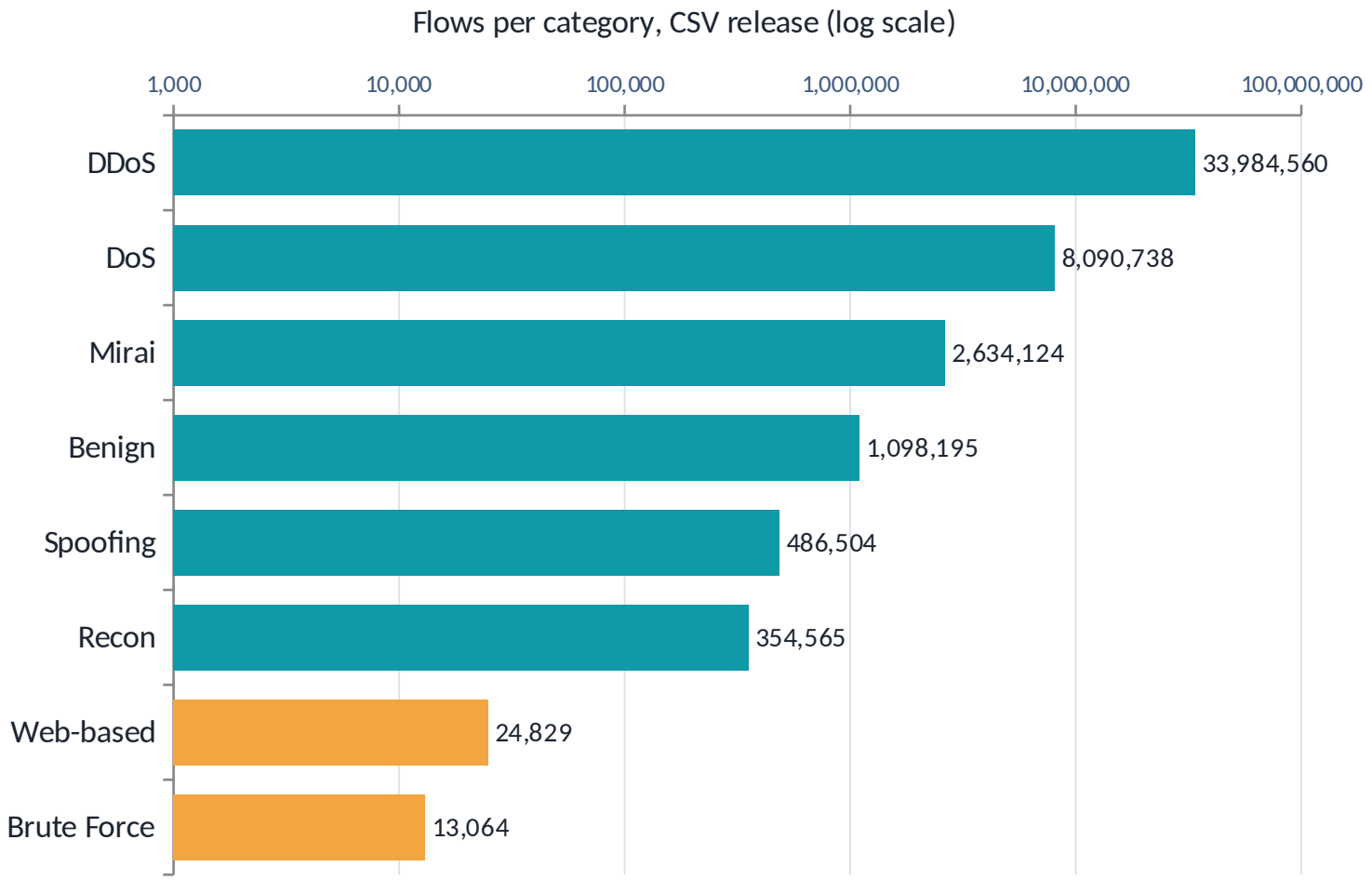
# CICIoT2023: the benchmark this project builds on

| 105 | 33 | 7 | 46 | ≈ 46.7 M |
|---|---|---|---|---|
| real IoT devices | attacks | attack categories | flow features | labelled flows |

**Neto et al., Sensors 2023 (Canadian Institute for Cybersecurity, UNB)**

- Real traffic captured in a lab topology of smart-home and IoT devices, attacked by malicious IoT devices.
- Categories: DDoS, DoS, Mirai, Reconnaissance, Spoofing, Web-based, Brute Force, plus benign traffic.
- Benchmarks machine-learning and deep-learning models at 2-class, 8-class and 34-class granularity.
- Listed in our course brief under “IoT Analytics Using Published IoT Datasets”.

# The gap: two attack categories are under 0.1% of the data




## Web + Brute Force

≈ 38,000 of ≈ 46.7 million flows (0.08%). A model trained on this data sees almost no examples of them.

### The question

Can we improve detection of these rare attacks without collecting new data, and without hurting the common ones?

# Three research questions, each with a measurable target

## RQ1 · Reproduce

REPRODUCTION

Does a tree-model reproduction (Random Forest, plus Decision Tree, LightGBM and Balanced Random Forest) match the paper's CICIoT2023 benchmark?

*Target: results in the paper's range, with every gap explained.*

## RQ2 · Rebalance

IMPROVEMENT

Does handling the imbalance (SMOTE, Borderline-SMOTE, SMOTE + undersampling, class weights, threshold tuning) raise recall on the rare classes?

*Target: higher rare-class recall and macro-F1 than the baseline on the same test set, confirmed by significance tests.*

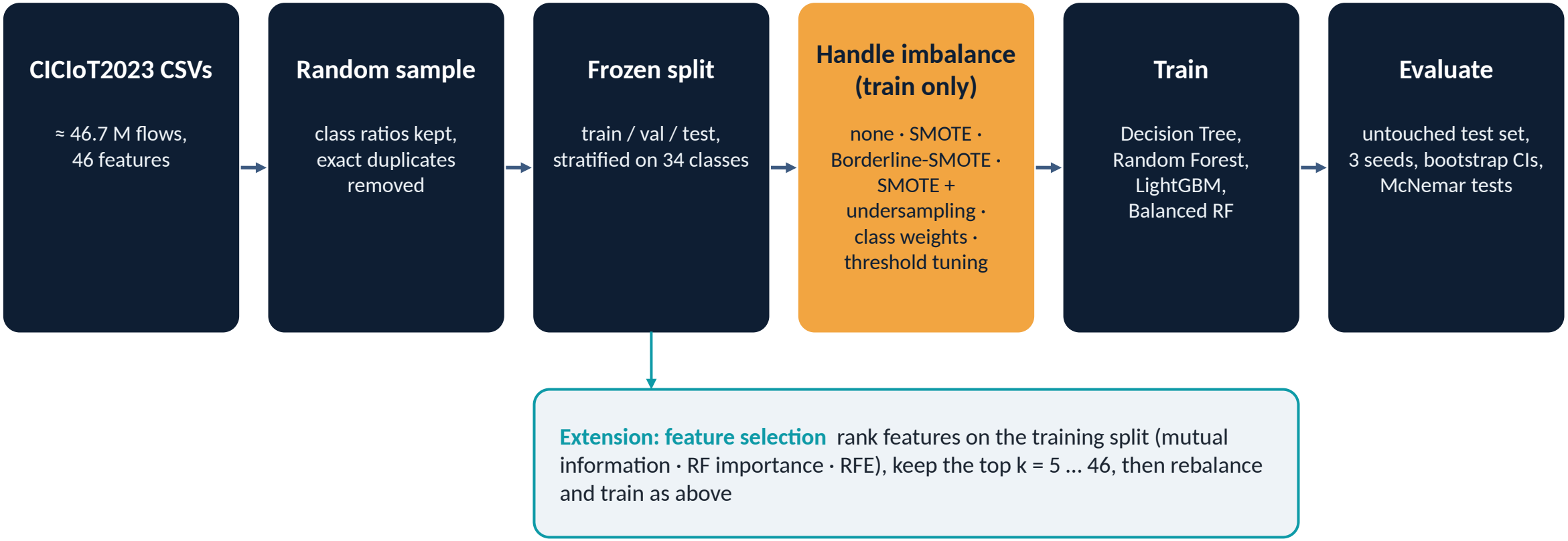
## RQ3 · Reduce

EXTENSION

Can a small feature subset (mutual information, RFE) keep that gain at lower cost?

*Target: ≤ 15 of 46 features keep most of the gain, with lower latency and model size.*

# Methodology: one pipeline, six ways to handle imbalance




**Leakage guard-rails:** the test set is split off first and never resampled; scaling, SMOTE and feature ranking are fit on training data only; settings are chosen on the validation split.

# What this project adds beyond the base paper

| | Base paper (benchmark) | This project |
|---|---|---|
| **Headline metric** | Overall accuracy and averaged scores | ✓ Per-class recall and macro-F1, rare classes reported separately |
| **Class imbalance** | Imbalanced data used as-is | ✓ Five imbalance methods compared head-to-head with the baseline |
| **Models** | Classical ML and deep-learning baselines | ✓ Tree models, including LightGBM and Balanced Random Forest |
| **Feature set** | All 46 features | ✓ Ranked subsets (MI, RF importance, RFE), from 5 to 46 features |
| **Deployment cost** | Not measured | ✓ Latency per flow, model size and training time for every model |
| **Rigour** | Benchmark scores reported | ✓ Duplicates removed, 3 seeds, bootstrap CIs, McNemar tests |

# How success will be measured

### Macro-F1

Every class counts equally, so rare classes can’t hide.

### Per-class recall

Share of each attack type actually caught; the core result.

### Rare-class recall

Mean recall over classes below 1% of the training data.

### Balanced accuracy, MCC

Imbalance-robust overall scores.

### Prediction latency

Microseconds per flow, for gateway deployment.

### Model size

Megabytes on disk, and tree node count.

**Fair comparison:** every model is scored on the same frozen, never-resampled, duplicate-free test set; mean ± standard deviation over 3 random seeds; every gain checked against the baseline with paired bootstrap 95% confidence intervals and McNemar tests.

# Feasible and already de-risked

**Built**

pipeline implemented and tested end-to-end

**< 12 h**

full experiment run in one free Kaggle CPU session

**₹ 0**

no special hardware, public dataset

| Risk | Mitigation |
| --- | --- |
| **Dataset too large (≈ 13 GB)** | Stream the files and keep a random sample with the true class ratios. |
| **SMOTE runs out of memory** | Raise rare classes to a capped target, not to the DDoS count; hybrid undersampling. |
| **Plain SMOTE gives no gain** | Four more methods (Borderline-SMOTE, hybrid, class weights, threshold tuning) are compared. |
| **Numbers differ from the paper** | Expected with a sample; the key comparison is baseline vs rebalanced on the same split. |

# Fits the course brief directly

### Project topic 4: IoT Intrusion Detection System

- DoS: DoS and DDoS categories
- Port scanning: Reconnaissance category
- Unauthorized access: Brute Force category
- Replay attacks are not in CICIoT2023; noted as a limitation

### Reading list: IoT Analytics Using Published IoT Datasets

- Base paper is item 3 (Neto et al., CICIoT2023)
- Related work draws on the security-framework list: MQTT IDS with deep learning and with LightGBM

### Links to the gateway and edge topics

- The extension targets models small and fast enough for an IoT gateway or edge AI server (topics 3 and 10)
- Latency and model size are measured, not assumed

# Plan and deliverables

### 1 · Reproduce

Sample, de-duplicate, split; baseline DT, RF, LightGBM, Balanced RF at 2, 8, 34 classes

### 2 · Improve

Five imbalance methods × three models vs baseline, 3 seeds, significance tests

### 3 · Extend

Feature ranking and top-k sweep: accuracy vs latency and size

### 4 · Report

Analysis, report, final presentation and demo

**GitHub repo**

code, tests, results

**Report**

PDF write-up

**Presentation**

results and demo

**Kaggle notebook**

one-click reproduction

# Approval requested

*Reproduce the CICIoT2023 benchmark, fix its blind spot on rare attacks, and make the fix light enough for an IoT gateway.*

### Clear problem

≈ 2,600 : 1 imbalance hides rare attacks

### Measurable outcome

rare-class recall and macro-F1 vs a reproduced baseline

### Low risk

pipeline built, public data, free compute

**References**
[1] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, A. A. Ghorbani, "CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment," Sensors, 23(13):5941, 2023. doi:10.3390/s23135941
[2] N. V. Chawla, K. W. Bowyer, L. O. Hall, W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," JAIR, 16:321–357, 2002.